L.11. Anticipating Risk In Science and Engineering

Definitions of Risk

Risk as an Unwanted Event

What it is?

Risk, seen as an unwanted event, means that something bad can happen, but it does not have to happen. The focus is on the bad event itself and the uncertainty around it. A situation involves risk when there is a chance that a harmful or negative event will occur. This way of understanding risk is common in everyday life and in simple risk analysis.

Example(s):

- Damage to the lungs from breathing in smoke is a big risk in fires.
- A chemical leak from a factory is a risk for people living nearby.
- Losing important files is a risk if computers are not backed up.
- Buildings collapsing is a risk during earthquakes.

Risk as a Cause of an Event

What it is?

Risk, understood as a cause of an unwanted event, refers to something that increases the chance that a bad event will happen. The focus is not on the bad event itself, but on the factor that can lead to it. In this view, a risk is a condition, action, or exposure that can cause harm. If the cause is present, the unwanted event becomes more likely.

Example(s):

- Smoking is a health risk because it can cause diseases like cancer.

- Driving too fast is a risk because it can lead to accidents.
- Using weak passwords is a risk because it can cause data theft.
- Working without safety equipment is a risk because it can lead to injuries.

Risk as Spread of Outcomes

What it is?

Risk, understood as the spread of outcomes, refers to how much possible results can vary. The focus is not only on bad outcomes, but on the fact that results can be very different from each other. In this view, a situation is more risky when the possible outcomes range from very good to very bad, and when it is hard to know which outcome will occur. This way of thinking about risk is common in finance and investments.

Example(s):

- A stock investment is risky because it can lead to high gains or large losses.
- Starting a new business is risky because it might succeed greatly or fail completely.
- Investing in cryptocurrency is risky due to large price changes.
- Choosing a variable interest loan is risky because payments can increase or decrease a lot.

Quantitative Notions of Risk

What it is?

Risk, in a quantitative sense, is defined using numbers. It can mean how likely an unwanted event is to happen, or how serious the consequences would be if it happens. Sometimes risk is described only by probability, and sometimes only by severity. In other cases, both are considered together. This way of defining risk is common in engineering, science, and risk analysis because it allows comparison and calculation.

Example(s):

- There is a 5% risk that a machine will fail during operation.
- The risk of flooding is high because the damage would be very severe.
- A medical treatment has a low risk of side effects, meaning the chance is small.
- An industrial accident is considered a high risk because it could cause many injuries.

Qualitative Notions of Risk

What it is?

Qualitative risk means describing risk using words instead of numbers. Risks are judged as high, medium, or low based on experience, expert judgement, or general understanding. This approach is useful when there is not enough data to calculate probabilities or when risks are hard to measure exactly.

Example(s):

- Working at great heights is considered a high risk job.
- Walking alone at night in an unsafe area is a high risk situation.
- A small software bug is seen as a low risk problem.
- Using new and untested technology is often described as a medium or high risk.

Decision Making Under Different Conditions

Decision-Making Under Certainty

What it is?

Decision-making under certainty means that there is only one possible outcome for an action. If you choose a certain action, you know exactly what will happen. There is no risk or uncertainty because the result is fixed and known in advance.

Example(s):

- If you press a light switch, the light turns on.
- Paying a fixed price for a bus ticket means you know the exact cost.
- Setting an alarm for 7:00 means it will ring at that time.
- Buying a product with a fixed price gives a known result: you pay that price and get the item.

Decision-Making Under Risk

What it is?

Decision-making under risk means that an action can lead to different outcomes, but the chances of each outcome are known. You do not know which outcome will happen, but you know how likely each one is. This allows decisions to be made using probabilities.

Example(s):

- Rolling a dice, where each number has a known chance of appearing.
- Playing a lottery, where the chance of winning is known.
- An insurance company knowing the probability of car accidents.
- Choosing a medical treatment when the success rate is known.

Decision-Making Under Ignorance

What it is?

Decision-making under ignorance means that an action can lead to several possible outcomes, but the chances of these outcomes are not known. You know what might happen, but you do not know how likely each outcome is. Because of this, it is harder to compare options using numbers.

Example(s):

- Launching a completely new product in an unknown market.
- Exploring a new technology that has never been tested before.
- Investing in a new company with no past data.

- Making decisions during a natural disaster when little information is available.

Decision-Making Under Deep Uncertainty

What it is?

Decision-making under deep uncertainty means that we do not know all possible outcomes, and we also do not know their probabilities. Some outcomes may be completely unexpected. This happens in complex or new situations where knowledge is very limited. In these cases, normal risk calculations do not work well.

Example(s):

- Making long-term climate policy decisions.
- Developing new technologies with unknown future effects.
- Planning for pandemics before they occur.
- Managing nuclear waste over very long time periods.

Expected Value Models of Risk

What it is?

Expected value models describe risk using numbers for both parts of an outcome: how likely it is to happen and how serious it would be if it happens. Each possible outcome has a probability and a level of damage or benefit. These models combine probability and severity to compare different choices and decide which option has the lowest or highest expected result.

Example(s):

- An insurance company calculates expected losses by combining accident probability with repair costs.

- A company compares projects by looking at expected profit and expected loss.
- A government estimates expected damage from floods using chance and damage size.
- A casino calculates expected winnings and losses for different games.

Decision Strategies

Principle of Insufficient Reason

What it is?

The principle of insufficient reason says that if we do not know how likely different outcomes are, we should treat them as equally likely. When there is no good reason to think one outcome is more likely than another, we assign the same probability to each. This is a simple way to make decisions when probability information is missing.

Example(s):

- If a box could contain one of four prizes and there is no information about which one it is, each prize is given a 25% chance.
- When choosing between two unknown options with no data, each is treated as equally likely.
- If a coin is found and nothing is known about it, one may assume heads and tails are equally likely.

Minimax

What it is?

Minimax is a decision rule used when outcomes are uncertain. It means choosing the action where the worst possible result is better than the worst results of the other options. The focus is on avoiding the most

serious loss, even if this means giving up the chance of a very good outcome.

Example(s):

- Choosing a job with a stable but lower salary instead of a risky job that could fail badly.
- Building a strong dam to reduce the worst possible flood damage, even if floods are unlikely.
- Saving money in a bank account instead of investing in a risky stock market.

Safety Factors

What it is?

Safety factors are used in engineering to make systems stronger than they need to be under normal conditions. Engineers add extra strength or capacity so the system can handle unexpected loads, stress, or errors. This helps reduce the chance of failure when something goes wrong or when conditions are worse than expected.

Example(s):

- A bridge is built to hold much more weight than normal traffic requires.
- A building is designed to withstand stronger winds than usually expected.
- An elevator cable is made much stronger than the maximum load it will carry.
- Aircraft parts are designed to handle stresses beyond normal use.

Precautionary Principle

What it is?

The precautionary principle says that if an action or policy might cause serious harm, we should be careful, even if there is no clear scientific agreement yet. When there is uncertainty, the responsibility is on those who want to take the action to show that it is safe. The idea is to prevent harm before it happens, rather than reacting after damage is done.

Example(s):

- Restricting a new chemical until it is shown to be safe for humans and the environment.
- Limiting the use of new technology when its long-term effects are unknown.
- Stopping the release of a new drug until enough testing proves it is not harmful.
- Regulating genetically modified organisms until their risks are better understood.

Subjective Evaluations of Risk

What it is?

Subjective evaluation of risk means that people judge risk based on their own feelings, values, and preferences, not only on numbers. People are often risk-averse, meaning they prefer safer options even if a risky option has a higher expected value. The St. Petersburg paradox shows that people do not always follow expected monetary value when making decisions. Daniel Bernoulli explained this by using *utility* instead of money, meaning that extra money matters less when you already have a lot. Risk judgments are also affected by whether a risk feels voluntary, familiar, or under personal control.

Example(s):

- Many people avoid gambling even if the expected gain is positive.
- People feel less worried about driving a car than flying, even though flying is statistically safer.

- Someone may reject a risky investment because they dislike uncertainty, even if the numbers look good.
- A person may accept more risk in activities they choose freely, like extreme sports, than in risks imposed on them, like pollution.

Utilitarian Approaches to Risk (Also known as Actualist Approach)

What it is?

In utilitarianism, actions are judged by their outcomes and by how much good or harm they produce. When outcomes are uncertain, it is not obvious how actions should be evaluated. Two main approaches are often discussed: the **actualist approach** and the **expected utility approach**.

According to the **actualist approach**, an action is judged only by the outcome that actually happens. Possible outcomes that did not occur do not matter. If the final result is good, the action is considered right, even if there was a high risk of a bad outcome.

According to the **expected utility approach**, an action is judged by its expected utility. This means that all possible outcomes are considered, together with how likely they are and how good or bad they would be. An action is morally right if it has the highest expected utility, even if the actual outcome later turns out to be bad.

The actualist approach is controversial because it makes moral judgment depend on luck. It ignores the risks that a person knowingly accepted. The expected utility approach avoids this problem by focusing on the decision itself and the information available at the time, rather than on the final outcome alone.

Example(s):

- You decide not to reinforce a bridge even though there is a 50% chance it will collapse. You send a very heavy transport over it, and luckily the bridge does not collapse. According to the actualist approach, your decision was right because the outcome was good. According to the

expected utility approach, the decision was wrong because you accepted a very high risk of serious harm.

- A doctor chooses a risky treatment with a low chance of success. If the patient survives, the actualist approach says the decision was right. The expected utility approach judges the decision based on the success rate and possible harm, not on the lucky outcome.
- A company ignores safety warnings and no accident happens. The actualist approach sees this as acceptable, while the expected utility approach may judge it as wrong because the expected harm was high.

Deontological Approach to Risk

What it is?

The deontological approach says that actions should be judged by rules and rights, not only by their outcomes. People have rights that must not be violated, even if violating them could lead to good results. When applied to risk, this raises the question of whether exposing someone to a risk of harm already violates their rights, even if the harm never actually happens.

In this view, even a small chance of serious harm can be morally wrong if it violates someone's rights. However, how rights are understood can depend on the situation. Important factors include whether the risk is taken voluntarily, whether the exposure to risk is intentional or unintentional, and whether the person exposed to the risk also benefits from it.

Example(s):

- Exposing people to dangerous pollution without their consent may violate their rights, even if no one becomes ill.
- A person who chooses to take part in a risky sport accepts the risk voluntarily, which may make it less of a rights violation.
- Forcing workers to take serious risks without proper protection can be wrong, even if no accidents occur.

- Medical experiments without informed consent are considered wrong, even if they lead to useful results.
